

MACHINEPULSE · DESIGN DECISIONS REGISTER

MachinePulse design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/machinepulse		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	Value engineering: saving that reused an existing USB charger removed (conflicts with the certified adapter, D2, and R17)
0.2	2026-10-02	Amish Chadha	Amish approved the recommendations for all six open decisions (2026-10-02); MPL-DDR-003 accepted; moved to decisions made
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; open decisions from MPL-DDR-001 to 003 and the review note

MachinePulse design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The current transformer's 3.5 mm plug passes the M16 gland's seal	Otherwise the plug must be fitted after the lead is through, or a larger gland used	MPL-DDR-003, P6
2	The pot magnets' M6 studs are at least 5 mm long and can be trimmed	The stud and the set screw share the block's left hole	MPL-DDR-003, P1, P2
3	The silicone grommet fits a 24 mm hole in 2.5 mm panel and grips a 20 mm boss	It seals the floor and must stay soft so it carries little vibration or heat	MPL-DDR-003, P4
4	The stock box: four corner pillars about 7 mm across, 2.5 mm walls, continuous use at 70 °C	The gland heights, the carrier position and the thermal model assume them	MPL-DDR-003, P7; MPL-CAL-001, G
5	The controller board: about 22 x 51 mm, two 20-pin rows 19.4 mm apart, no octal PSRAM	The carrier's header sockets are spaced for it	MPL-DDR-003, P5; MPL-CAL-001, G4
6	The light pipe's hole size (3.2 mm clearance or 3.0 mm press fit) and its length (about 29 mm)	The lid hole and the LED position follow from it	MPL-DDR-003, P10
7	The probe clip's disc magnets are rated 150 °C or more	The clip sits on frames up to 100 °C	MPL-DDR-003, P9
8	The accelerometer breakout's size (about 18 x 18 mm) and that its back can be bonded flat	It is bonded to the 20 mm boss top	MPL-DDR-003, P8

■ Value engineering

Value-engineering target: USD 81 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 83.50 (USD 2.50 over the target). Main cost drivers and savings worth trying:

- The largest lines are the accelerometer (USD 15, the least certain price), the current transformer (USD 10), the controller (USD 9), the box with its glands (USD 9.50) and the power adapter (USD 8).
- Making the design constructable added USD 2.50: the third gland and the larger M16 gland (USD 0.50), the carrier board's sockets, terminals and light pipe (USD 1.00), the made probe clip with its magnets in place of a bought clip (USD 0.50 net) and the grommet, set screw and carrier fixings (USD 0.50).

- Savings worth trying: the steel pads (USD 4 for two) are needed only on aluminium-framed machines and could be left out of the standard kit; the accelerometer price should be checked against current breakout boards; an adapter and cable bought in quantity would cover the gap; a certified adapter is kept for every kit (D2, R17).

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D10: Wi-Fi first, certified 5 V adapter, IIS3DWB-class sensor, one CT on one phase, first-week baseline alerts, TwinKit or any MQTT broker, stock ABS box on 5 mm nylon stand-offs, makerspace pilot, no budget change, features every minute and a spectrum every 15 minutes	Amish: "i accept all your recommendations, go with them across all repos."	MPL-DDR-001, MPL-DDR-002
2026-09-25	N1 high-temperature pot magnets rated 120 °C; N2 R8 restated (2 °C to 85 °C, 4 °C from 85 to 100 °C); N3 keep the R3 and R5 targets and measure first at TRL 4	Amish, same instruction	MPL-DDR-002
2026-09-30	Build plans are written in the approved format, and the design is made physically buildable as the pictures are drawn	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	MPL-DDR-003 (changes accepted on 2026-10-02, below)
2026-09-30	Open decisions are kept in this register, not in the build plan	Amish: "don't log outstanding decisions in this build plan - that is not the place for it."	This register
2026-10-01	<code>budget_usd</code> is a value-engineering target, not a limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens."	This register, Value engineering
2026-10-02	Design for construction accepted: all ten changes of Table 1 and their knock-on changes, as made	Amish: "i approve your recommendations for all 555 open decisions."	MPL-DDR-003, Table 1
2026-10-02	Power entry: (a) for the prototype, the cut USB cable through an M12 gland to a screw terminal	Amish: "i approve your recommendations for all 555 open decisions."	MPL-DDR-003, A1
2026-10-02	Status button: (a) for the prototype, on the carrier board and pressed with the lid off; a sealed lid button only if field pairing shows it is needed	Amish: "i approve your recommendations for all 555 open decisions."	MPL-DDR-003, A2; review note 2026-09-26, item 1
2026-10-02	R4: restated as a relative energy trend for the first build, with any kWh figure reported as an estimate; the voltage reference (c) stays an option for sites that need absolute energy	Amish: "i approve your recommendations for all 555 open decisions."	MPL-DDR-001, O2; MPL-DDR-002
2026-10-02	Co-design partner for alerts and dashboard: the makerspace pilot already decided, with its shop lead deciding what "changed" should mean, and the maintenance person of one small production machine shop as a second voice (first candidate type to approach, not agreed)	Amish: "i approve your recommendations for all 555 open decisions."	MPL-DDR-001, O1
2026-10-02			

Date	Decision	Decided by	Record
	Enclosure colour: a single-colour stock box is bought; two-tone stays a later option	Amish: "i approve your recommendations for all 555 open decisions."	Review note 2026-09-26, item 2